

The Lorenz Attractor:

Chaos and Climate

Calum Heldt

University of Nebraska

✉ calum.heldt@gmail.com

@ calheldt.xyz

Abstract

In this project, we will describe how the Lorenz system behaves for different parameter values, narrowing in on one which dictates the stability of the famous “butterfly attractor”. We will discuss this attractor’s basin of attraction and fractal dimension, then computationally demonstrate its chaos.

Finally, we will discuss weather patterns and how to derive a more accurate prediction despite the Lorenz system’s inherent chaos. Then, we will provide some examples of how to apply these results to study climate.

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1 Introduction and Motivation

This project examines the Lorenz system (1963):

$$\begin{cases} \frac{dx}{dt} = \sigma(y - x), \\ \frac{dy}{dt} = \rho x - xz - y, \\ \frac{dz}{dt} = xy - \beta z, \end{cases} \quad (1)$$

where $\sigma, \rho, \beta > 0$.

This was originally derived from the Rayleigh-Bénard convection experiment, which describes the motion in a fluid heated below. Maybe surprisingly, convection models were more recently applied to study atmospheric turbulence [Pad78], which is quite fitting because Lorenz himself preferred to study atmospheric motion [FGM83]. This project approaches the Lorenz system from the atmospheric perspective when possible, first taking time to analyze the system's behavior.

Namely, we would like to know when solutions to (1) reach an equilibrium, what kind of equilibria it achieves, and when these solutions exhibit chaotic properties. Additionally, we will demonstrate chaos, so we will also discuss how to more accurately predict solutions to (1). Finally, we will comment on some applications of (1) to climate modeling.

1.1 Physical Interpretation

The meaning of each variable and parameter is often obscured behind the pretense of several different physical experiments resulting in models similar to (1). Here, we try to distill them using [KE13]. First, x, y , and z do not represent a spatial coordinate, but rather a spatial average. Namely, (1) x is a spatial average of *hydrodynamic velocity*, the velocity of particles in a given fluid, (2) y is a spatial average of temperature, and (3) z is a spatial average of temperature *gradient*, the rate of change of temperature.

Next, (1) σ is related to the *Prandtl number* of the fluid, which itself relates viscosity, or momentum transport, to conductivity, or heat transport [Rap17], (2) ρ is related to the *Rayleigh number* of the fluid, which determines the impact of convection caused by buoyancy [ZNKN21], and β is related to the *aspect ratio* of the given domain, approximately the height divided by width [Rap17]. Specific values used come from experiments, often keeping σ and β fixed, as they would given a specific fluid inside of a domain, while letting ρ vary to determine its effect.

2 Equilibria

This section explores the equilibria of (1) and their stability. To find an equilibrium point of (1), each time derivative is 0. First, $\frac{dx}{dt} = 0$, so $x = y$. Making this substitution, the next conditions simplify to

$$\begin{aligned}\rho x - xz - x &= x(\rho - 1 - z) = 0, \\ x^2 - \beta z &= 0.\end{aligned}$$

Clearly $x = y = 0$ is a solution to the former equation, and the latter then implies $z = 0$. Otherwise, we divide the former by $x \neq 0$ and get $z = \rho - 1$, so $x^2 = \beta(\rho - 1)$ and thus $x = y = \pm\sqrt{\beta(\rho - 1)}$. Hence nontrivial equilibrium solutions are only defined when $\rho \geq 1$.

Comment.

Notice that this implies there exists one equilibrium for $\rho < 1$, two equilibria for $\rho = 1$, and three equilibria for $\rho > 1$. Hence $\rho = 1$ is the bifurcation value for equilibria. This will be discussed more in Section 2.3.

Now we have the equilibria $(x, y, z) = (0, 0, 0)$ and $(\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1)$, provided $\rho \geq 1$.

2.1 Linearizing About Equilibria

Here, we use the Jacobian operator to linearize about the different equilibria. We will treat $(0, 0, 0)$ separately from the pair $(\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1)$. First, we can calculate the Jacobian of the system, denoted A :

$$A = \begin{bmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{bmatrix}.$$

2.1.1 Equilibrium at $(0, 0, 0)$

At $(x, y, z) = (0, 0, 0)$, the Jacobian equals

$$A = \begin{bmatrix} -\sigma & \sigma & 0 \\ \rho & -1 & 0 \\ 0 & 0 & -\beta \end{bmatrix}.$$

To find eigenvalues λ , we solve for roots of the cubic polynomial

$$\begin{aligned}\det(A - \lambda I) &= \begin{vmatrix} -\sigma - \lambda & \sigma & 0 \\ \rho & -1 - \lambda & 0 \\ 0 & 0 & -\beta - \lambda \end{vmatrix} \\ &= (-\sigma - \lambda)(-1 - \lambda)(-\beta - \lambda) - \sigma\rho(-\beta - \lambda) \\ &= -(\beta + \lambda)\left[(\sigma + \lambda)(1 + \lambda) - \sigma\rho\right] \\ &= -(\beta + \lambda)\left[\lambda^2 + (\sigma + 1)\lambda + \sigma(1 - \rho)\right].\end{aligned}$$

Clearly $\lambda = -\beta$ is a root. Using the quadratic formula, the other two roots are

$$\lambda = \frac{-(\sigma + 1) \pm \sqrt{(\sigma + 1)^2 - 4\sigma(1 - \rho)}}{2}.$$

2.1.2 Equilibrium at $(\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1)$

Recall that the Jacobian is

$$A = \begin{bmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{bmatrix}.$$

At these new equilibria, this simplifies to

$$A = \begin{bmatrix} -\sigma & \sigma & 0 \\ 1 & -1 & \mp\sqrt{\beta(\rho - 1)} \\ \pm\sqrt{\beta(\rho - 1)} & \pm\sqrt{\beta(\rho - 1)} & -\beta \end{bmatrix}.$$

To find eigenvalues λ , we can solve for roots of the cubic polynomial

$$\begin{aligned}\det(A - \lambda I) &= \begin{vmatrix} -\sigma - \lambda & \sigma & 0 \\ 1 & -1 - \lambda & \mp\sqrt{\beta(\rho - 1)} \\ \pm\sqrt{\beta(\rho - 1)} & \pm\sqrt{\beta(\rho - 1)} & -\beta - \lambda \end{vmatrix} \\ &= (-\sigma - \lambda)\left((-1 - \lambda)(-\beta - \lambda) + \beta(\rho - 1)\right) - \sigma\left((-\beta - \lambda) + \beta(\rho - 1)\right) \\ &= -(\sigma + \lambda)\left(\lambda^2 + (1 + \beta)\lambda + \beta\rho\right) - \sigma\lambda - 2\sigma\beta - \sigma\beta\rho \\ &= -\lambda^3 - (1 + \beta + \sigma)\lambda^2 + (-\sigma\beta - \beta\rho)\lambda - \sigma\beta - 2\sigma\beta\rho \\ &= -\left[\lambda^3 + (\sigma + 1 + \beta)\lambda^2 + \beta(\sigma + \rho)\lambda + 2\sigma\beta(\rho - 1)\right].\end{aligned}$$

Flipping the sign, this is the same as solving for solutions to

$$0 = \lambda^3 + (\sigma + 1 + \beta)\lambda^2 + \beta(\sigma + \rho)\lambda + 2\sigma\beta(\rho - 1),$$

If we denote these solutions $\lambda_1, \lambda_2, \lambda_3$, then they must also satisfy

$$\begin{aligned}0 &= (\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) \\ &= \lambda^3 - (\lambda_1 + \lambda_2 + \lambda_3)\lambda^2 + (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1)\lambda - \lambda_1\lambda_2\lambda_3,\end{aligned}$$

which derives a system of equations that the eigenvalues must satisfy.

Notice that the sign of each eigenvalue for a given equilibrium is relevant to determine its stability, which is discussed more in terms of the stability bifurcations in (2.3). However, we can even circumvent eigenvalue discussions by applying a stability theorem.

2.2 Finding a Lyapunov Function

Now we can determine an appropriate Lyapunov Function for the equilibria, and use it to apply relevant theorems.

Theorem 2.1.

(Lyapunov Stability) Let x_* be an equilibrium point for (1). Let $L : \mathcal{O} \rightarrow \mathbb{R}$ be a differentiable function defined on an open set $\mathcal{O}$ containing x_* . Suppose further that

g) $L(x_*) = 0$ and $L(x) > 0$ if $x \neq x_*$, and

g) $L(x) \leq 0$ in $\mathcal{O} \setminus \{x_*\}$,

then x_* is stable. Furthermore, if L also satisfies

c) $L(x) < 0$ in $\mathcal{O} \setminus \{x_*\}$,

then x_* is asymptotically stable [HSD13].

Define the function

$$L(x, y, z) := \frac{1}{\sigma}x^2 + y^2 + z^2.$$

Clearly $L(x, y, z) \geq 0$ for all $(x, y, z) \in \mathbb{R}$. Suppose (x, y, z) is a solution to the equations from (1), then

$$\begin{aligned} \frac{1}{2} \frac{dL}{dt} &= x \frac{1}{\sigma} \frac{dx}{dt} + y \frac{dy}{dt} + z \frac{dz}{dt} \\ &= x(y - x) + y(\rho x - xz - y) + z(xy - \beta z) \\ &= xy - x^2 + \rho xy - xyz - y^2 + xyz - \beta z^2 \\ &= -x^2 + (1 + \rho)xy - y^2 - \beta z^2 \\ &= -\left(x^2 - (1 + \rho)xy + \frac{(1+\rho)^2}{4}y^2\right) + \left(1 - \frac{(1+\rho)^2}{4}\right)y^2 - \beta z^2 \\ &= -\left(x - \frac{1+\rho}{2}y\right)^2 + \left(\frac{(1+\rho)^2}{4} - 1\right)y^2 - \beta z^2. \end{aligned}$$

When $\frac{(1+\rho)^2}{4} - 1 \leq 0$, equivalently when $\rho \leq 1$, $\frac{dL}{dt}$ is never positive. Further, if we suppose $(x, y, z) \neq (0, 0, 0)$, then $\frac{dL}{dt}$ is strictly negative when $\rho < 1$. For larger values of ρ , $\frac{dL}{dt}$ does not necessarily exceed 0. By Theorem 2.1, $(0, 0, 0)$ is not only stable but *asymptotically* stable when $\rho < 1$. Again, this points toward the bifurcation of ρ , so we have narrowed down a parameter of interest to discuss more in the following section.

2.3 Stability Bifurcations

We have already explored equilibria for general parameter values. We now explore the bifurcations of ρ specifically, which as we have already discovered will influence the amount and stability of equilibria.

2.3.1 Equilibrium at $(0, 0, 0)$

At $(x, y, z) = (0, 0, 0)$, recall from Section 2.1.1 that the eigenvalues are

$$-\beta, \frac{-(\sigma + 1) \pm \sqrt{(\sigma + 1)^2 - 4\sigma(1 - \rho)}}{2}.$$

Case 1: When $0 < \rho < 1$, we already know $\sigma > 1$, so $(\sigma + 1)^2 - 4\sigma(1 - \rho) < (\sigma + 1)^2 - 4\sigma = (\sigma - 1)^2$. Then the largest eigenvalue from the quadratic equation would satisfy

$$\lambda < \frac{-(\sigma + 1) + (\sigma - 1)}{2} = -1,$$

so all eigenvalues are negative, and thus the equilibrium $(0, 0, 0)$ is stable.

Case 2: However, when $\rho > 1$, $(\sigma + 1)^2 - 4\sigma(1 - \rho) > (\sigma + 1)^2$, so the largest eigenvalue from the quadratic equation satisfies

$$\lambda > \frac{-(\sigma + 1) + (\sigma + 1)}{2} = 0.$$

In this case, at least one eigenvalue is positive, so the equilibrium $(0, 0, 0)$ is unstable.

2.3.2 Equilibrium at $(\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1)$

At $(x, y, z) = (\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1)$, recall from Section 2.1.2 that the eigenvalues are given by the system of equations

$$\begin{cases} -(\lambda_1 + \lambda_2 + \lambda_3) = \sigma + 1 + \beta, \\ \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1 = \beta(\sigma + \rho), \\ -\lambda_1\lambda_2\lambda_3 = 2\sigma\beta(\rho - 1). \end{cases} \quad (2)$$

Case 1: Suppose $\rho = 1$. From (2), $\lambda_1\lambda_2\lambda_3 = 2\sigma\beta(\rho - 1) = 0$, so at least one eigenvalue is 0. Without loss of generality, call this eigenvalue λ_1 . Then we must also have

$$\begin{aligned} 0 &= \lambda^2 + (\sigma + 1 + \beta)\lambda + \beta(\sigma + 1) \\ &= (\lambda - \lambda_2)(\lambda - \lambda_3) \\ &= \lambda^2 - (\lambda_2 + \lambda_3)\lambda + \lambda_2\lambda_3. \end{aligned} \quad (3)$$

First we can show that λ_2, λ_3 are both real. Notice that the quadratic formula applied to (3) gives us

$$\lambda_2, \lambda_3 = \frac{1}{2} \left(-(\sigma + 1 + \beta) \pm \sqrt{(\sigma + 1 + \beta)^2 - 4\beta(\sigma + 1)} \right).$$

The determinant equals $(\sigma + 1 - \beta)^2 \geq 0$, so $\lambda_2, \lambda_3 \in \mathbb{R}$. Now, since $\sigma, \beta, \rho > 0$, $\lambda_2 \lambda_3 = \beta(\sigma + \rho) > 0$, so the real parts of λ_2 and λ_3 have the same sign. If they're both positive, then $-(\lambda_2 + \lambda_3) = \sigma + 1 + \beta < 0$, which is a contradiction. Hence λ_2 and λ_3 must both be negative.

Case 2: Next, suppose $\rho := \rho_0 = 1 + \epsilon$ for some small $\epsilon > 0$. Since the roots of any polynomial change continuously with respect to changes in any of the constants, we can denote $\lambda := \lambda_1(\epsilon)$ to be the eigenvalue that equals 0 when $\epsilon = 0$ and is small when ϵ is small. Namely, we choose ϵ small enough so that both λ_2 and λ_3 remain negative. From (2), $-\lambda_1 \lambda_2 \lambda_3 = 2\sigma\beta\epsilon > 0$, so either one or three eigenvalues are negative. Since at least two eigenvalues (λ_2 and λ_3) are negative, then λ_1 must also be negative.

In experiments, the exact value of ϵ need not be known, but we can at least derive a rough estimate. Since λ_1 is small, we can assume $\lambda_1^3 \approx 0$, so again the quadratic formula applied to (3) gives us

$$\begin{aligned} \lambda_1 &\approx \frac{1}{2} \left(-(\sigma + 1 + \beta) \pm \sqrt{(\sigma + 1 + \beta)^2 - 4\beta(\sigma + 1 + \epsilon)} \right) \\ &\approx \frac{1}{2} \left(-(\sigma + 1 + \beta) \pm \sqrt{(\sigma + 1 - \beta)^2 - 4\epsilon} \right). \end{aligned} \quad (4)$$

At this point, we can theoretically find λ_1 in terms of ϵ . But if we further assume that $\lambda_1^2 \approx 0$, we are left with the linear equation

$$\begin{aligned} \beta(\sigma + 1 + \epsilon)\lambda_1 + 2\sigma\beta\epsilon &\approx 0, \\ \lambda_1 &\approx -2\sigma\epsilon/(\sigma + 1 + \epsilon) \end{aligned}$$

Of course then we can also bound ϵ in terms of λ_1 :

$$\begin{aligned} \lambda_1(\sigma + 1) + \lambda_1\epsilon &\approx -2\sigma\epsilon \\ \lambda_1(\sigma + 1) &\approx -\epsilon(2\sigma + \lambda_1) \\ \lambda_1(\sigma + 1) &\approx -\epsilon(2\sigma + \lambda_1) \\ \epsilon &\approx -\lambda_1(\sigma + 1)/(2\sigma + \lambda_1). \end{aligned} \quad (5)$$

Now $\epsilon > 0$ and $\lambda_1 < 0$, so we need also that $2\sigma + \lambda_1 > 0$, so $0 > \lambda_1 > -2\sigma$. In other words, we already know λ_1 is small, so we just need to make sure σ isn't too small.

Again, the eigenvalues depend continuously on initial data, so instability can only occur if the points cross either the real or imaginary axes. We already know that $\epsilon = 0$ results in one zero and two negative eigenvalues, and ϵ small enough results in all negative eigenvalues. By continuity, these roots likely won't *cross* the real axis, so we can assume they cross the imaginary axis for an instability.

Case 3: Now, we want to find the exact ρ large enough that two of these eigenvalues become complex conjugates with zero real part, satisfying the above comment about instability. Working backwards, we can suppose we have $\lambda_2 = ai$, $\lambda_3 = -ai$, and again $\lambda_1 < 0$.

Then solutions to (2) become solutions to

$$\begin{aligned} 0 &= (\lambda - \lambda_1)(\lambda - ai)(\lambda + ai) = (\lambda - \lambda_1)(\lambda^2 + a^2) \\ &= \lambda^3 - \lambda_1\lambda^2 + a^2\lambda - a^2\lambda_1. \end{aligned}$$

Looking at (2), we have the equalities

$$\begin{cases} \sigma + 1 + \beta = -\lambda_1, \\ \beta(\sigma + \rho) = a^2, \\ 2\sigma\beta(\rho - 1) = -a^2\lambda_1. \end{cases} \quad (6)$$

If we consider σ and β to be given, this is a system of three equations with three unknowns that we can solve by clearing λ_1 and a^2 from the third equation and reducing as follows:

$$\begin{aligned} 2\sigma\beta(\rho - 1) &= \beta(\sigma + \rho)(\sigma + 1 + \beta) \\ 2\sigma\rho - 2\sigma &= \sigma(\sigma + 1 + \beta) + \rho(\sigma + 1 + \beta) \\ 2\sigma\rho - \rho(\sigma + 1 + \beta) &= \sigma(\sigma + 1 + \beta) + 2\sigma \\ \rho(2\sigma - (\sigma + 1 + \beta)) &= \sigma((\sigma + 1 + \beta) + 2) \\ \rho(\sigma - 1 - \beta) &= \sigma(\sigma + \beta + 3) \\ \rho &= \sigma(\sigma + \beta + 3)/(\sigma - 1 - \beta) \end{aligned} \quad (7)$$

so at $\rho := \rho_1 = \sigma(\sigma + \beta + 3)/(\sigma - 1 - \beta)$, the solutions are complex conjugates with zero real part, as desired. Now for any $\rho_0 < \rho < \rho_1$, the roots must all be negative. Otherwise, there would be some discontinuity in the roots as a function of ϵ , so $\rho = \rho_1$ is the exact critical value. This also means that we require $1 < \rho_1$, or if we know ϵ , then $1 + \epsilon < \rho_1$.

2.3.3 Examples

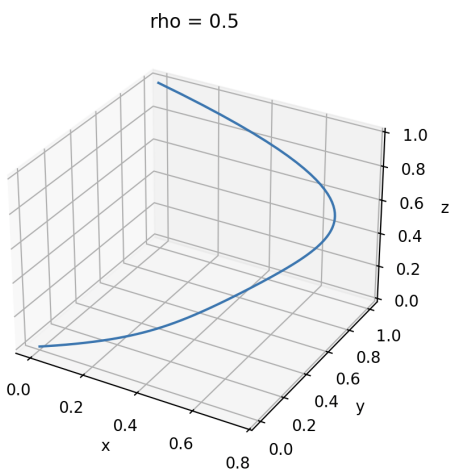
Now we have thoroughly examined each of the equilibria and the factors influencing their stability. Namely, 3 critical bifurcation values emerge at $\rho = 1, \rho_0$, and ρ_1 as defined above, yielding 5 intervals of interest.

We choose the classical values $\sigma = 10$ and $\beta = 8/3$, and further suppose $\lambda_1 \cong 0.1$ so that our preliminary estimate for ϵ holds, then (5) implies

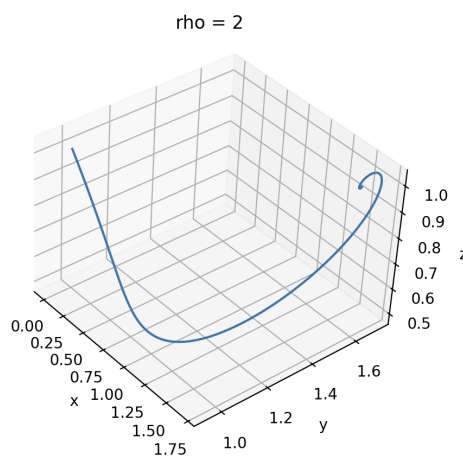
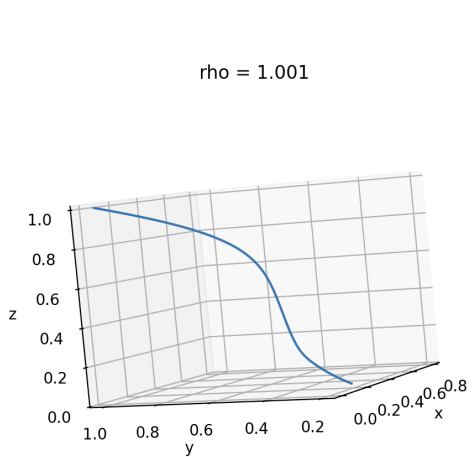
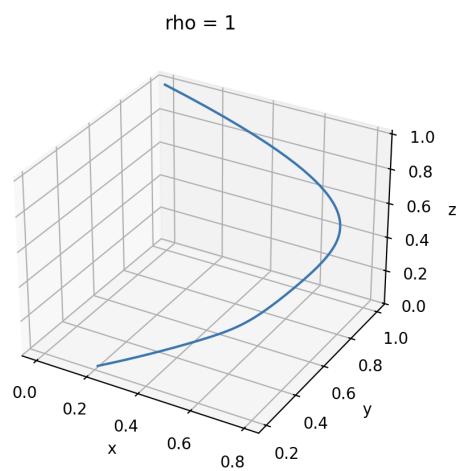
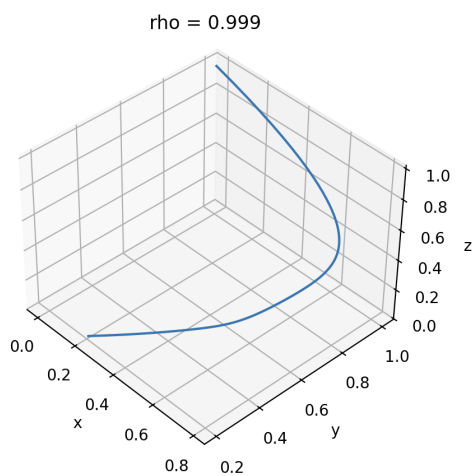
$$\rho_0 = 1 + \epsilon \approx 1 + \lambda_1(\sigma + 1)/(2\sigma + \lambda_1) \approx 1.05.$$

Also if we choose $\sigma = 10$ and $\beta = 8/3$, then (7) implies $\rho_1 = \frac{470}{19} \approx 24.74$. This gives us a good picture of the 5 approximate intervals of interest. Using some randomly generated, representative initial conditions, we can graph solutions corresponding with each interval.

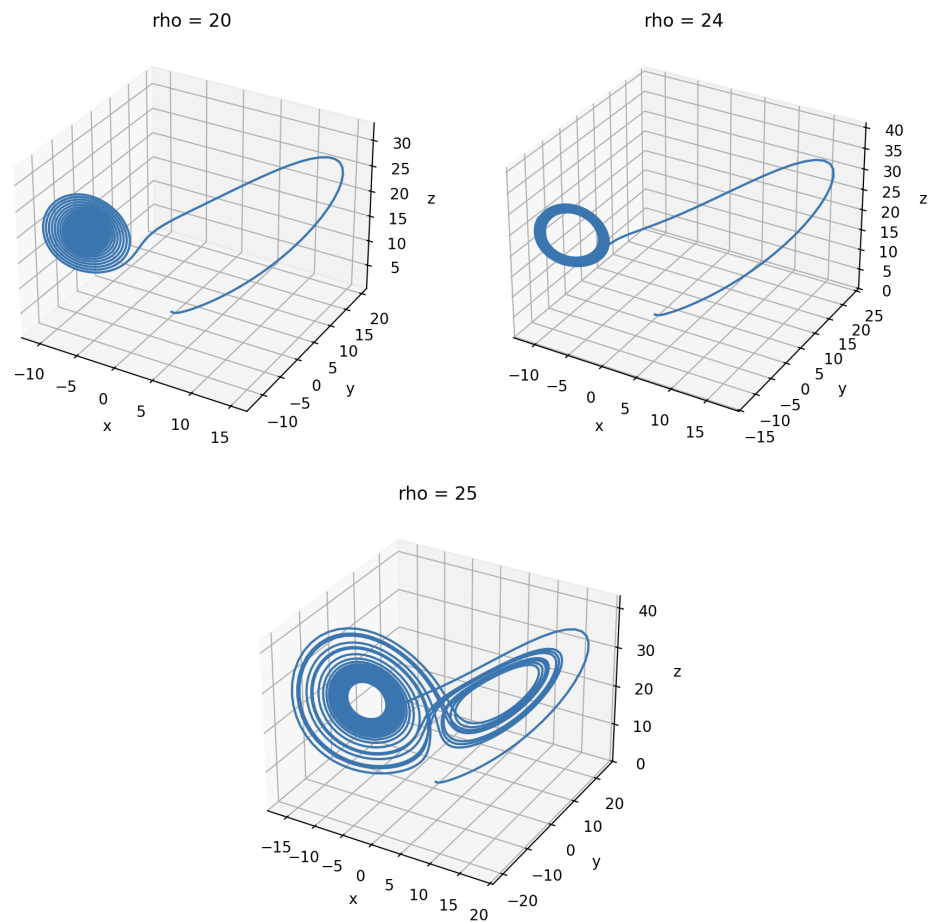
a) **For** $0 < \rho < 1$, we graph $\rho = 0.5$:



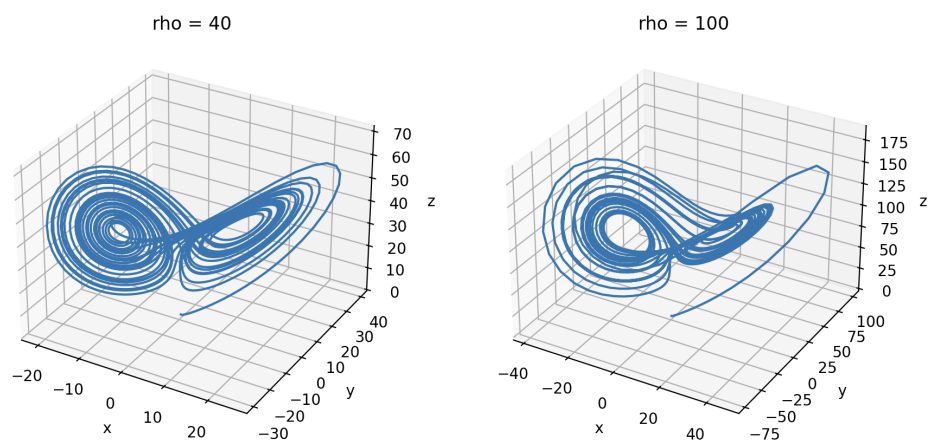
b) For $\rho \approx 1$, we graph $\rho \in \{0.999, 1, 1.001\}$:

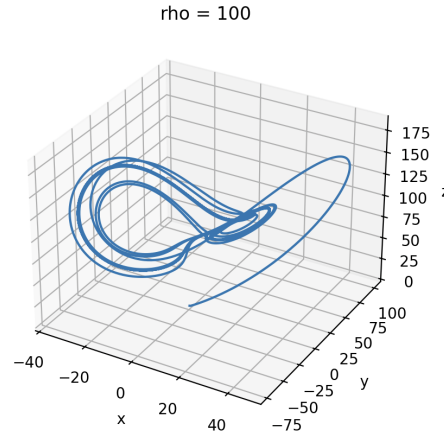


c) For $\rho \approx \rho_1$, we graph $\rho \in \{20, 24, 25\}$:

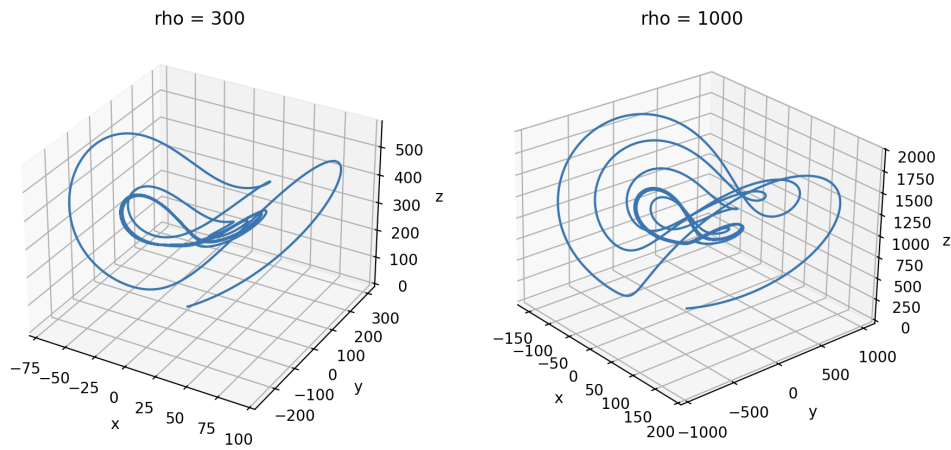


d) For larger ρ , we graph $\rho \in \{40, 100, 300\}$:





Here, I changed my time step from 0.01 to 0.001 to accommodate $\rho = 100$ suddenly looking like too coarse of an estimate.



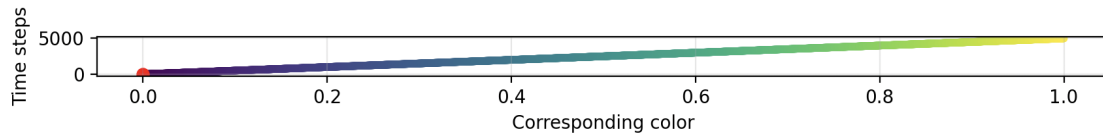
This backs up the ρ values found, where there's just one arc for small enough ρ , which starts to curve into itself and eventually converges to an equilibrium around $\rho = 1$. Then, this equilibrium splits into two around $\rho = \rho_1$. This corroborates what we found analytically, namely:

- a) For $0 < \rho < 1$, the origin is both stable and the only equilibrium.
- b) For $1 < \rho < \rho_0$, the origin becomes unstable and the pair of equilibria is now defined and both are stable.
- c) For $\rho_0 < \rho < \rho_1$, there could be potentially chaotic orbits.
- d) For $\rho \cong \rho_1$, an attractor known as the “butterfly attractor” emerges, which we will now discuss in Section 3.
- e) For the chosen values of $\rho \gg \rho_1$, we see similar behavior.

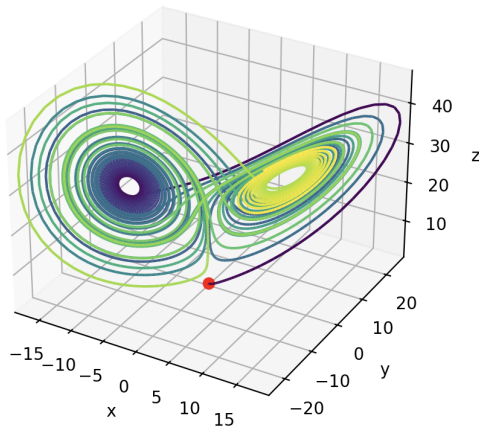
3 The Butterfly Attractor

Recall the pair of equilibria $(\pm\sqrt{\beta(\rho-1)}, \pm\sqrt{\beta(\rho-1)}, \rho-1)$ which exist (and are distinct from the origin) only for $\rho > 1$. They compose the so-called *butterfly* attractor, in which trajectories approach both equilibria in the pair as time goes to infinity. First, let's get a better picture of this attractor.

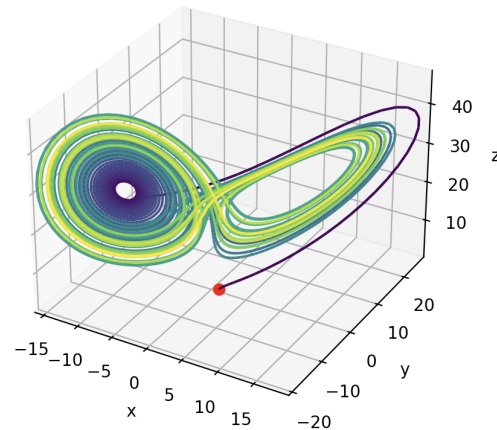
3.1 Graphing the Butterfly Attractor



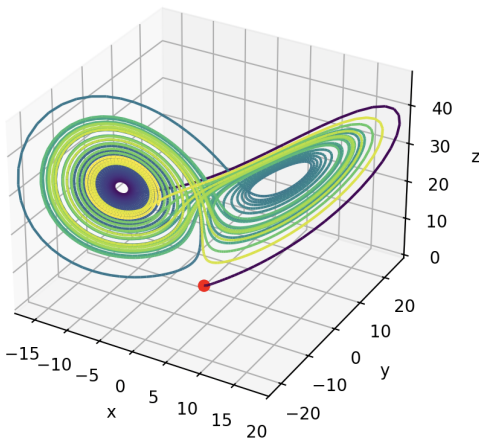
(a) Time mapping



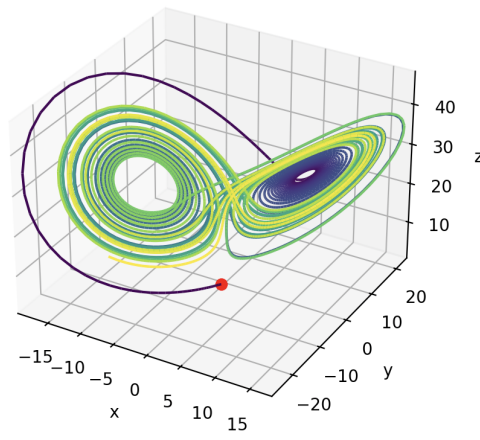
(b) $(-0.10, 0.35, 0.75)$



(c) $(0.80, 0.69, 0.14)$



(d) $(-0.07, 0.93, -0.91)$



(e) $(-0.25, -0.82, 0.58)$

Figure 1: The Lorenz system with different initial conditions

These graphs are the output of a Runge-Kutta 4 time-step solver in Python, where the solution to (1) after $5 \cdot 10^4$ steps of size 10^{-2} . The Python codes that I wrote are found in

Sections 7.1 and 7.2. Included in Figure 1 are some results for different initial conditions, assuming the classical parameters chosen to display the butterfly attractor: $\beta = 8/3$, $\sigma = 10$, and $\rho = 28$.

Comment.

Notice that each of the initial conditions graphed are within a very small region, yet have very different plots. This suggests high sensitivity to initial conditions, one of the requirements for chaos, which will be discussed more in Section 4.

3.2 Basins of Attraction

We will prove that solutions to the system (1) are bounded, and actually this bound describes an *absorbing set*, which then generates the attractor. First, define the bounding function

$$B(x, y, z) := \rho x^2 + \sigma y^2 + \sigma(z - 2\rho)^2.$$

Notice that for a solution (x, y, z) of (1),

$$\begin{aligned} \frac{1}{2} \frac{dB}{dt} &= x \frac{dx}{dt} + \sigma y \frac{dy}{dt} + \sigma(z - 2\rho) \frac{dz}{dt} \\ &= x(\sigma(y - x)) + \sigma y(\rho x - xz - y) + \sigma(z - 2\rho)(xy - \beta z) \\ &= \sigma xy - \sigma x^2 + \sigma \rho xy - \sigma xyz - \sigma y^2 + \sigma xy z - \sigma \beta z^2 - 2\sigma \rho xy + 2\sigma \rho \beta z \\ &= \sigma(-x^2 + (1 - \rho)xy - y^2 - \beta z^2 + 2\rho \beta z) \\ &= -\sigma\left(x - \frac{1-\rho}{2}\right)^2 - \sigma\left(1 - \frac{(1-\rho)^2}{4}\right)y^2 - \sigma\beta(z^2 - 2\rho z) \\ &= -\sigma\left(x - \frac{1-\rho}{2}\right)^2 - \sigma\left(1 - \frac{(1-\rho)^2}{4}\right)y^2 - \sigma\beta(z - \rho)^2 + \sigma\beta\rho^2. \end{aligned} \tag{8}$$

If we set $\frac{dB}{dt} \leq 0$, we get the equation of $\mathbb{R}^3$ minus a filled-in ellipsoid in the (x, y, z) plane, whose boundary is given by the equation

$$E(x, y, z) := -\sigma\left(x - \frac{1-\rho}{2}\right)^2 - \sigma\left(1 - \frac{(1-\rho)^2}{4}\right)y^2 - \sigma\beta(z - \rho)^2 + \sigma\beta\rho^2, \tag{9}$$

This ellipsoid is centered at $(\frac{1-\rho}{2}, 0, \rho)$ and has x , y , and z semi-axes of

$$\left(\frac{1}{\beta\rho^2}\right)^{1/2}, \quad \left(\frac{4}{4\beta\rho^2 - \beta\rho^2(1-\rho)^2}\right)^{1/2}, \quad \text{and} \quad \left(\frac{1}{\rho^2}\right)^{1/2},$$

respectively. Hence solutions outside of this ellipsoid eventually converge to a trajectory inside of the closed and bounded ellipsoid, given by $E := \{\overline{\mathbb{R}^3 \setminus E(x, y, z)} : (x, y, z) \text{ solves (1)}\}$, so we call this ellipsoid an *absorbing set*.

Definition 1.

The *global attractor* is the ω -limit set of a closed and bounded absorbing set [Bab06].

We have already seen this attractor, but now we can concretely define it as $\omega(E)$.

Definition 2.

The ω -limit set of a set X is defined as

$$\omega(X) = \bigcap_{t \geq 0} \overline{\bigcup_{s \geq t} S(s)X},$$

where $S(\cdot)$ is a semi-group operator given by $S(t)x := \phi_t(x)$, the trajectory of the system starting at x going until time t , and $S(t)X := \{S(s)x : x \in X\}$ [Hai08].

So, $\omega(E) = \bigcap_{t \geq 0} \overline{\bigcup_{s \geq t} S(s)X}$, which can be found computationally using similar code as in Section 3.1.

3.3 Fractal Dimension of the Butterfly Attractor

For an enjoyable tangent, this section describes previous work about the fractal dimension of the butterfly attractor in [Vis04]. This paper uses a certain generalization of the Jacobian matrix, which describes change at a point. This generalization is called the *monodromy* (or *Floquet*) matrix, which measures change over an orbit.

Definition 3.

Specifically, the *monodromy* matrix M satisfies $\phi_{t+T} = M(T)\phi_t$ for all points in a period- T closed orbit [Iri06].

Also, each matrix N such that $e^{TN} = M(T)$ defines a period- T function $t \rightarrow P(t)$ such that $\phi_t = P(t)e^{tN}$ for all $t \in \mathbb{R}$.

Definition 4.

A *characteristic multiplier* is an eigenvalue of e^{TN} , which are the same regardless of choice of matrix N [Chi99].

Definition 5.

A *Floquet exponent* is an eigenvalue of e^{TN} , which are the same regardless of choice of matrix N [Chi99].

This ties back nicely into our earlier class discussions, because a Lyapunov exponent is simply the real part of a Floquet exponent. Now, we can finally find the Hausdorff dimension of (1). Suppose F is a period- T , invertible mapping of $\mathbb{R}^3$ with a chaotic attractor. Namely, we can take a F to be a map from $\mathbb{R}^3$ to a collection of period- T points in the butterfly attractor of (1). Denote D_T to be the solution to

$$\sum_i \lambda_i^{D_T-1} = 1,$$

where each λ_i is a characteristic multiplier of F^T at some period- T point, x_i . Then [GOY88, ACE⁺87] argue that the sequence D_T converges to the Hausdorff dimension of the system (1) as $t \rightarrow \infty$.

In [Vis04], these numbers are computed from 4 to 20. The results determined that fractal dimension of this attractor is roughly 2.06. Eight decimal places are in fact computed, with an error of less than 10^{-5} . This gives us a pretty good idea of what this butterfly attractor looks like: at the two points, it is locally 2-dimensional. The “wings” on each of these points interact minimally, very slightly increasing the fractal dimension to roughly 2.06.

More perspective on this attractor is shown in Figure 2 below:

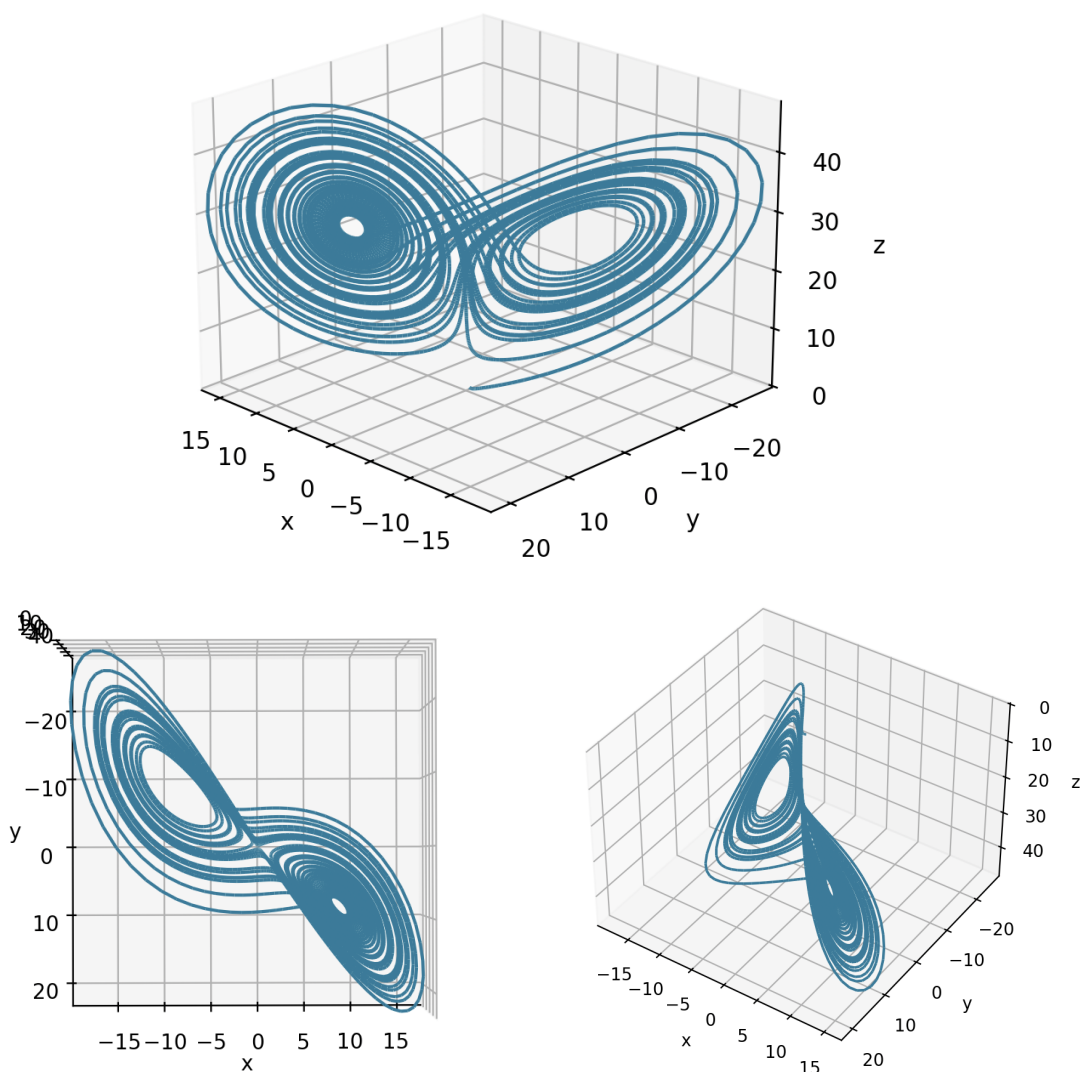


Figure 2: The Butterfly Attractor with different perspectives

4 Chaos

4.1 Finding Lyapunov Exponents

As hinted thus far, the butterfly attractor is indeed chaotic. To prove this computationally, we can find some positive Lyapunov exponents. The code for these computations is found in Section 7.3, which was adapted from the MATLAB code found in [TBD18]. Results are shown in Figure 3 below. Again, we take the classical parameters $\beta = 8/3$, $\sigma = 10$, and $\rho = 28$.

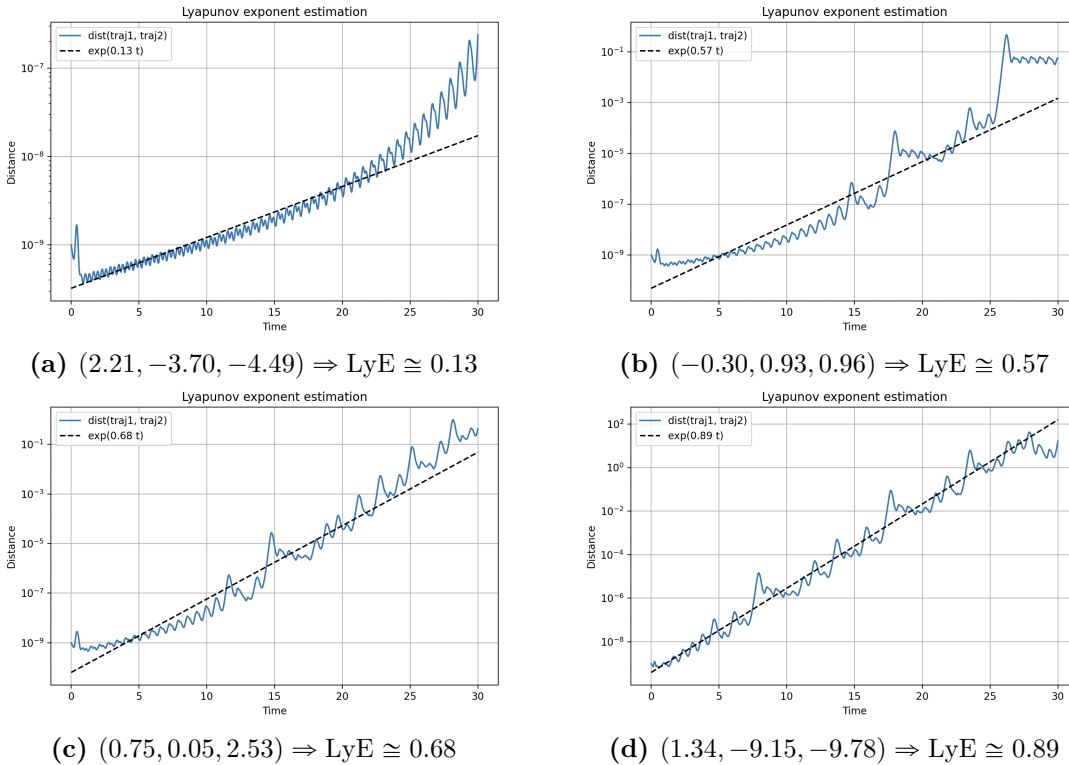


Figure 3: Lyapunov exponents of the Lorenz system

We see that the same points plotted in Figure 3 tend towards the butterfly attractor and have positive Lyapunov exponent, so the butterfly attractor itself exhibits chaos. In fact, the largest Lyapunov exponent of the attractor is around 0.905 [Vis98], which we have almost replicated in Figure 3(d).

Further, the Lyapunov exponents visual representations given in Figure 3 demonstrate that points infinitesimally close together separate exponentially fast. Hence the system satisfies one of the conditions for chaotic behavior, high sensitivity to initial conditions.

4.2 A Chaotic Attractor

Now that we have demonstrated one criterion of chaos, we need only show that there are initial points which exhibit aperiodic long-term behavior. We can arbitrarily choose $(2, 2, 2)$ and notice that, even after 10^4 time steps, there is still no clear pattern (Figure 4).

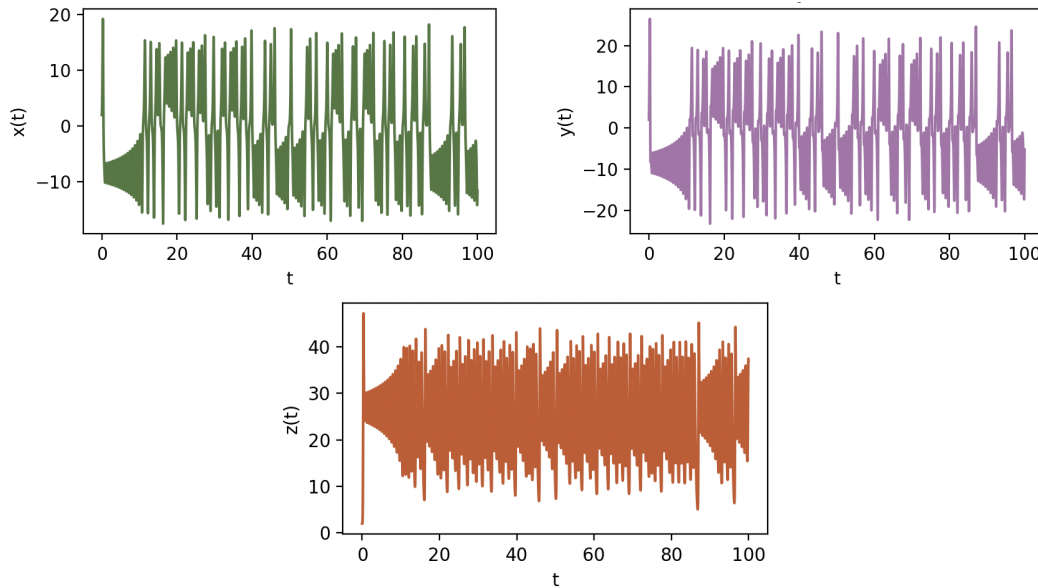


Figure 4: The Lorenz system in yz , xz , and xy planes

Hypothetically, this pattern could repeat in $2 \cdot 10^4$ time steps, but as chaos has already been proven [MM95], we can move on to the applications.

5 Climate Applications

As mentioned, atmospheric applications are more in the spirit of Lorenz’s own studies (as well as my own) so we will finally explore those further, giving purpose to the previous results.

5.1 Predicting Climate With Averages

For a long-term, aperiodic weather event – like global temperature – taking a longer running average provides a more accurate extrapolation of the given data [Shi21]. For example, using January-September monthly average temperatures will not result in a very good prediction for October’s average temperature. This is shown in the last several data points of Figure 5.

Instead, the previous several Octobers’ average temperatures would be a better data set to extrapolate from, but this method would also propagate errors because it only calls on data from much further back.

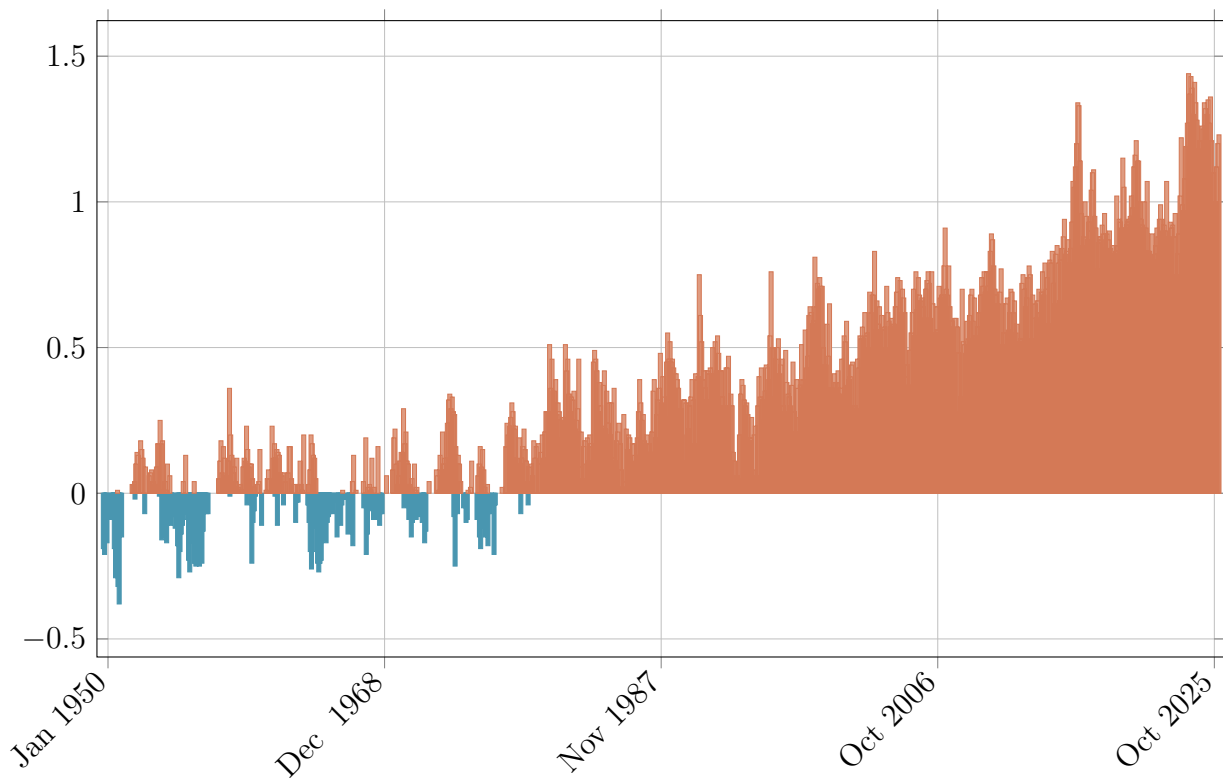


Figure 5: Global Land and Ocean Temperature Anomaly Monthly Average [fEI25]

One solution is to take an even longer running average because this provides a much smoother prediction, as in Figure 6. Additionally, we can see that with the same data set, there are no periodic orbits, so the data is not expected to return to a certain value. If we were to extrapolate this information to the proceeding months, we would also not see a return to some previous pattern. Hence the accuracy of this method is greater.

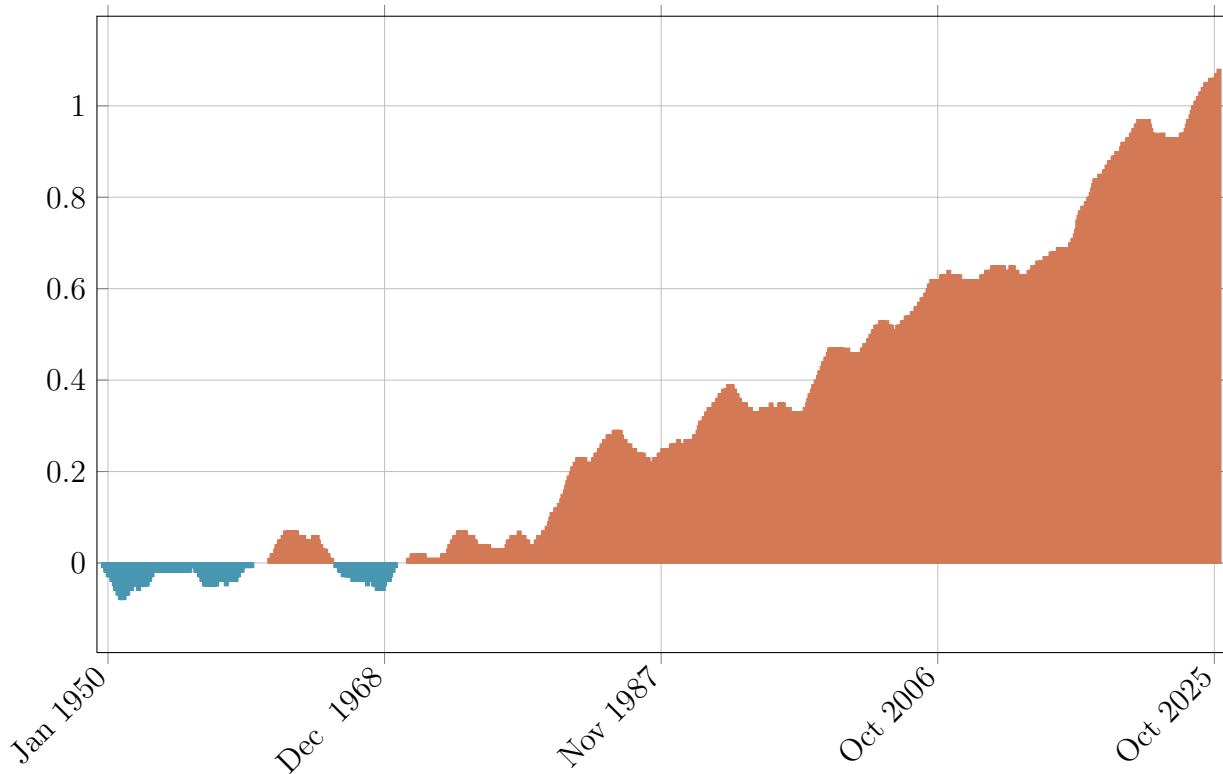


Figure 6: Global Land and Ocean Temperature Anomaly 5-Yearly Average [fEI25]

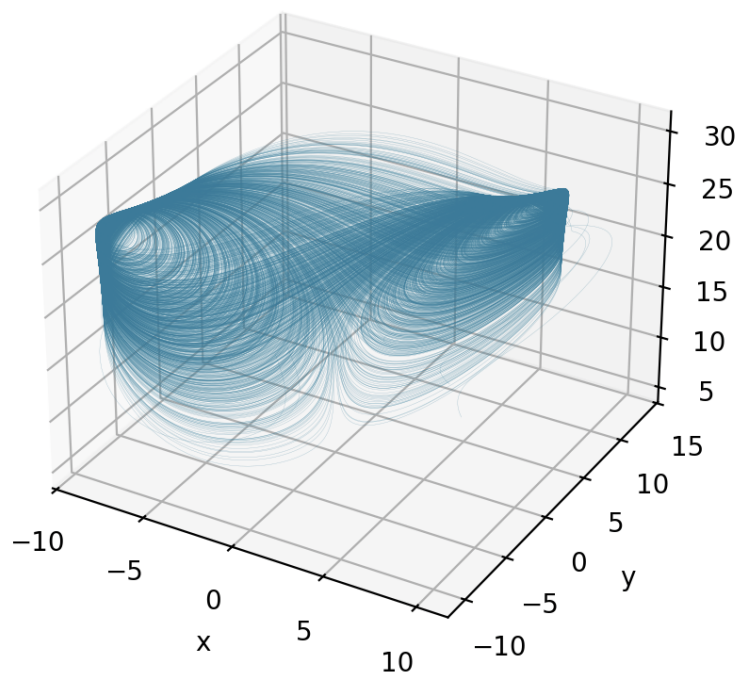
Moreover, the notion that systems become more predictable when taking a longer running average applies to chaotic systems, like (1). In this circumstance, we can take a running average over an increasing number of time steps.

5.2 Predicting the Lorenz System with Averages

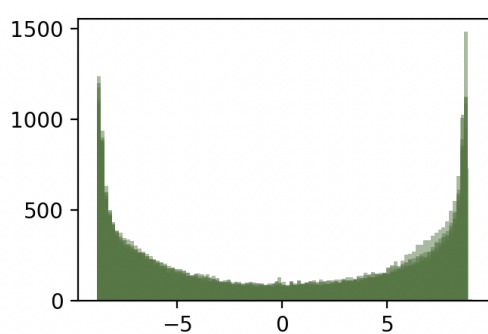
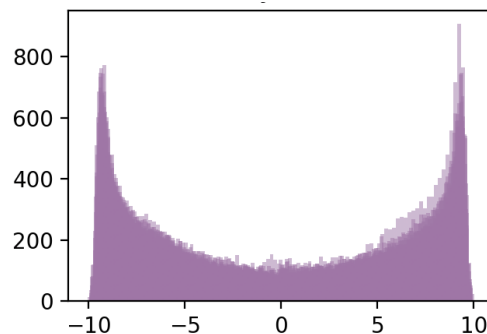
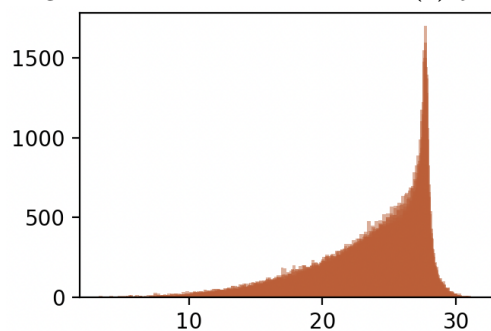
Now we know that we can better predict solutions of the Lorenz system when taking a long-term average, because this provides a sort of damping effect for the chaotic behavior. I recreated the results of [Shi21] using my own code (Section 7.4). Here, we not only dampen chaos in time by taking a running average, but also dampen its effects in space by overlaying the plots of several averaged solutions with initial conditions within a very small (about 10^{-8}) radius of each other. For 50 time steps, we get Figure 7 below.

As found in [Shi21], who offers the above results for a variety of running averages, taking a running average of solutions of the Lorenz system demonstrates two things:

- 1) In the x - and y - axes, the *amount* of regions whose initial points produce convergent solutions increases as the length of running average increases.
- 2) In the z -axis, the *area* of regions whose initial points produce convergent solutions decreases as the length of running average increases.



(a) All solutions

(b) x -axis histogram(c) y -axis histogram(d) z -axis histogram**Figure 7:** The Lorenz system running average

5.3 A Note on Climate Applications

Originally, I sought to detail applications of the Lorenz system to climate modeling. However, I found that this would not be entirely accurate. First, adding higher-order averages of x , y , and z are said to generate very different results, so whether or not the Lorenz system can be accurately applied to atmospheric modeling is up for debate [KE13].

However, many mathematicians have still worked on applying this model to the atmosphere. Namely, due to its chaotic nature, this system “qualitatively revealed the essence of a finite predictability within a chaotic system” [SPZZ23]. Despite this original gumption to employ (1) in studying climate, it has only been shown to predict atmospheric changes for around 2 weeks [Cha66]. Hence mathematicians have moved on to finding even better models.

6 Conclusion

Now we have described how the Lorenz system behaves for different parameter values, showing that σ and β have little affect on stability, while ρ has three critical bifurcation values that dictate the stability of each of 3 equilibria. From this, we discovered the butterfly attractor, which investigated further.

First, we plotted varying random or arbitrarily chosen initial conditions to demonstrate how points behaved around the attractor, derived a formula for the attractor's basin of attraction, and described previous results about its fractal dimension. Eventually, we gave a computational demonstration in the attractor's chaos, which all showed how interesting this attractor could be.

Finally, we discussed weather patterns, displaying how to derive a more accurate prediction despite the system's inherent chaos. Ultimately, we were able to draw some conclusions about weather patterns dictated by the Lorenz system of equations, showing that this system is not only interesting, but at least a little bit useful.

7 Python Code

7.1 Lorenz System

```

1  import matplotlib.pyplot as plt
2  import numpy as np
3  import random
4
5  # random floats
6  a = random.uniform(-1, 1)
7  b = random.uniform(-1, 1)
8  c = random.uniform(-1, 1)
9
10 # random initial value
11 y1_0 = np.array([a, b, c])
12
13 # constants
14 t0 = 0.0                                # initial time
15 T = 100                                # final time
16 dt = 0.01                             # step size
17 beta = 8/3; sigma = 10; rho = 20      # parameters
18
19 # lorenz system
20 lorenz = lambda t , x : np.array([
21     sigma*(x[1] - x[0]),
22     x[0]*(rho - x[2]) - x[1],
23     x[0]*x[1] - beta*x[2]
24 ])
25
26 # solve and plot solution
27 X , t = rk4_ndim(lorenz, y1_0, t0, T, dt, 3)
28
29 fig = plt.figure()
30 ax = fig.add_subplot(projection = '3d')
31 ax.plot(X[0,:], X[1,:], X[2,:])
32 ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')
33
34 plt.show()

```

7.2 Runge-Kutta 4 Solver

```

1  import numpy as np
2
3  def rk4_ndim(f, y0, t0, T, dt, n):
4
5      N = int(np.floor((T-t0)/dt) + 1) # fixed number of steps

```



```

6     t = t0+dt*np.arange(0,N)
7
8     y = np.zeros((n, N), dtype = float)
9     y[:, 0] = y0 # Set initial value
10
11    for n in range(0,N-1):
12
13        k1 = f(t[n], y[:, n])
14        k2 = f(t[n] + dt/2, y[:, n] + (dt*k1)/2)
15        k3 = f(t[n] + dt/2, y[:, n] + (dt*k2)/2)
16        k4 = f(t[n] + dt, y[:, n] + (dt*k3))
17
18        y[:, n+1] = y[:, n] + (dt/6)*(k1 + (2*k2) + (2*k3)+k4)
19
20    return y,t

```

7.3 Lyapunov Exponents

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3  import random
4
5  # random floats
6  a = random.uniform(-1, 1)
7  b = random.uniform(-1, 1)
8  c = random.uniform(-1, 1)
9
10 # random initial value
11 y1_0 = np.array([a, b, c])
12
13 # small initial perturbation
14 eps = 1e-9
15 y2_0 = y1_0 + np.array([0, 0, eps])
16
17 # constants
18 t0 = 0.0 # initial time
19 T = 30 # final time
20 dt = 0.01 # step size
21 beta = 8/3; sigma = 10; rho = 28 # parameters
22
23 # lorenz system
24 lorenz = lambda t, x : np.array([
25     sigma*(x[1] - x[0]),
26     x[0]*(rho - x[2]) - x[1],
27     x[0]*x[1] - beta*x[2]
28 ])

```

```

29
30 # solve for both trajectories
31 X1, t = rk4_ndim(lorenz, y1_0, t0, T, dt, 3)
32 X2, _ = rk4_ndim(lorenz, y2_0, t0, T, dt, 3)
33
34 # Distance
35 d = np.linalg.norm(X1.T-X2.T, axis=1) # take transpose for correct dimension
36
37 # linear fit given by log(distance) = slope * t + intercept
38 slope, intercept = np.polyfit(t[t <= 25], np.log(d[mask]), 1)
39
40 plt.figure(figsize=(8,5))
41 plt.semilogy(t, d, label="dist(traj1, traj2)")
42 plt.semilogy(t, np.exp(intercept+slope*t), 'k--', label=f'exp({slope:.2f}t)')
43 plt.xlabel("Time")
44 plt.ylabel("Distance")
45 plt.title("Lyapunov exponent estimation")
46 plt.legend()
47 plt.grid(True)
48 plt.show()

```

7.4 Average of Lorenz System

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3  from matplotlib.gridspec import GridSpec
4  import random
5
6  # constants
7  N = 50 # length of running avg
8  t0 = 0.0 # initial time
9  dt = .01 # step 0.1
10 num_steps = 20000 # number of steps
11 T = dt*num_steps # final time
12 beta = 8/3; sigma = 10; rho = 28 # parameters
13
14 # lorenz system
15 lorenz = lambda t, x : np.array([
16     sigma*(x[1] - x[0]),
17     x[0]*(rho - x[2]) - x[1],
18     x[0]*x[1] - beta*x[2]
19 ])
20
21 # setup plots
22 fig = plt.figure(figsize=(12, 8)) # 3x2, first row = twice height of bottom
23 gs = GridSpec(2, 3, height_ratios=[2, 1], figure=fig) # actual layout

```

```
24
25 # 3D
26 ax1 = fig.add_subplot(gs[0, :], projection='3d')
27 ax1.set_xlabel('x')
28 ax1.set_ylabel('y')
29 ax1.set_zlabel('z')
30 ax1.legend()
31
32 # x(t) histogram
33 ax2 = fig.add_subplot(gs[1, 0])=
34 ax2.set_title('x(t) Histogram')
35
36 # y(t) histogram
37 ax3 = fig.add_subplot(gs[1, 1])
38 ax3.set_title('y(t) Histogram')
39
40 # z(t) histogram
41 ax4 = fig.add_subplot(gs[1, 2])
42 ax4.set_title('z(t) Histogram')
43
44 # solve then take averages
45 for n in range(10):
46     # given initial condition
47     y0 = [1, 1.27, 14.132]
48
49     # random floats
50     a = 1^-8*random.uniform(-1, 1)
51     b = 1^-8*random.uniform(-1, 1)
52     c = 1^-8*random.uniform(-1, 1)
53
54     # random, small perturbation
55     y0 += np.array([a, b, c])
56
57     # solve lorenz given conditions
58     X , t = rk4_ndim(lorenz, y0, t0, T, dt, 3)
59
60     # convert solution to proper shape
61     X = np.asarray(X)
62     X = X.T
63
64     # take averages
65     sum = np.cumsum(X, axis=0)
66     avg = np.zeros_like(X)
67
68     for i in range(len(X)):
69         if i < N:
70             avg[i] = sum[i] / (i+1)
```

```
71         else:
72             avg[i] = (sum[i] - sum[i-N]) / N
73
74         # plot
75         ax1.plot(avg[:,0], avg[:,1], avg[:,2], alpha=0.5, linewidth=0.1)
76         ax2.hist(avg[:,0], bins=100, alpha=0.5, linewidth=0.1)
77         ax3.hist(avg[:,1], bins=100, alpha=0.5, linewidth=0.1)
78         ax4.hist(avg[:,2], bins=100, alpha=0.5, linewidth=0.1)
79
80 plt.show()
```

Comment.

For more up-to-date codes and a variety of other differential equations, please take a look at [my Github repository](#) ☺.

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